INTRODUCTION

Brain tumor detection with image processing

This system looks at a brain MRI scan, finds the tumor and suggests its type: glioma, meningioma or pituitary. It uses classic image processing only. There is no AI and no training.

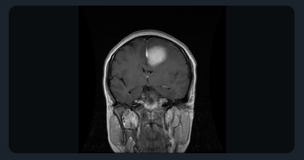
What it uses

- Python with NumPy, SciPy and scikit-image
- Filters to clean and sharpen the image
- Brightness thresholds to separate tissues
- Shape operations and measurements

How it avoids AI

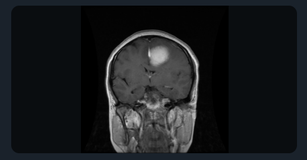
- No training data and no learned model
- Every decision is a measurement checked against a written rule
- The same scan always gives the same answer
- Each step can be shown and explained

How it works



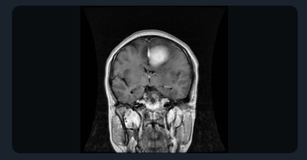
1 Load

Read the scan as grayscale



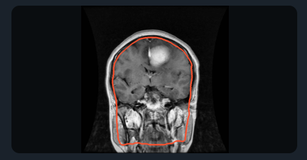
2 Denoise

Median filter removes speckle



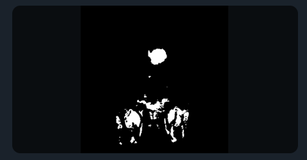
3 Enhance

CLAHE boosts local contrast



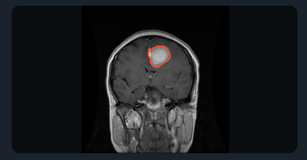
4 Remove skull

Threshold and erosion



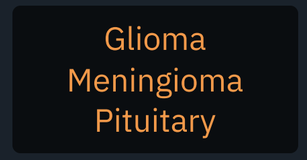
5 Find bright areas

Otsu brightness thresholds



6 Check rules

Size, shape and brightness



7 Suggest type

Rules on location and look